

Abdou Rakib Abente

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SUMMARY

AI and ML Engineer with hands-on experience building and deploying production AI systems including LLM-powered agents, RAG pipelines, and intelligent automation tools. Comfortable working across the full stack from model training to deployment. Self-directed builder who ships real systems used by real people, not just experiments. Proficient in Python, PyTorch, TensorFlow, HuggingFace, Claude API, Gemini API, and FastAPI.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Bash
AI and LLMs: Anthropic Claude API, Gemini 2.5 Flash, OpenAI API, Qwen2.5-7B, RAG pipelines, prompt engineering, tool calling, agentic orchestration
ML Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, XGBoost, LightGBM, SHAP, Optuna, Sentence Transformers
Specializations: Large Language Models, Generative AI, Agentic AI Systems, NLP, Computer Vision, Deep Learning, Embeddings, Semantic Search
Build and Deploy: FastAPI, Gradio, Streamlit, HuggingFace Spaces, ZeroGPU, Railway, AWS S3, Docker basics, GitHub Actions, Git
Data and Databases: SQLite, Pandas, NumPy, REST APIs, feature engineering, data preprocessing
Languages Spoken: English (Fluent), French (Fluent)

EXPERIENCE

Machine Learning Engineer Intern - Buildawn Labs

November 2025 to January 2026 | Remote

- Built and evaluated ML models for computer vision, NLP, and LLM-based applications integrated into production data pipelines via FastAPI.
- Developed data preprocessing and feature engineering pipelines for structured enterprise datasets; improved throughput by 30 percent.
- Collaborated with senior engineers and data scientists to research and implement modern AI solutions for real client use cases.

Direct Support Professional - Kenccid

2024 to Present | Philadelphia, PA

- Supported individuals in a government-funded social services program; maintained detailed documentation and coordinated with multidisciplinary teams.

PROJECTS

ML Research Assistant - Production RAG Chatbot

Qwen2.5-7B-Instruct, RAG, Sentence Transformers, Gradio, ZeroGPU | live: huggingface.co/spaces/Abdourakib/ml-research-assistant

- Built and deployed a production AI assistant for ML engineers to search research papers, find datasets and models, discover libraries, and get code generation help.
- Implemented a full RAG pipeline with semantic search, document chunking, embedding-based retrieval, and context-aware generation using Qwen2.5-7B.
- Deployed on HuggingFace Spaces with ZeroGPU; resolved real production issues including memory constraints, routing bugs, and version conflicts to maintain stable operation.

Autonomous Job Application Agent - Multi-Agent Workflow Automation

Anthropic Claude API, Python, JSearch API, SQLite, Gmail OAuth2, Streamlit, Railway | github.com/Abdrakib/job-agent

- Built an end-to-end agentic system that automates a full business workflow: discovers jobs via API, scores each one using Claude AI, generates tailored resumes and cover letters, and delivers results via Gmail.
- Designed multi-agent orchestration with tool calling, structured outputs, and SQLite-backed state management; deployed on Railway with APScheduler for daily automated runs.
- Built a Streamlit dashboard for monitoring application history, scores, and outcomes in real time.

Autonomous Trading System - Live AI-Powered Automation

Python, OANDA API, Anthropic Claude API, SQLite, Streamlit, Telegram Bot | github.com/Abdrakib/trading-system

- Built a live trading automation system with 5 concurrent AI strategy agents including Trend Following, Mean Reversion, Breakout, SMC Institutional, and News Fade.
- Integrated real-time OANDA API data feeds, dynamic risk management, market regime classification, and Telegram bot alerts; all trade results written to SQLite database.
- Deployed on DigitalOcean with a Streamlit dashboard for live monitoring of positions, P&L, and strategy performance.

Explainable AutoML Pipeline Agent

Qwen2.5-7B, Gradio, ZeroGPU, SHAP, Optuna, AWS S3 | huggingface.co/spaces/Abdourakib/explainable-ml-pipeline-analysis-agent

- Deployed an agentic ML pipeline that accepts raw tabular data and automates feature engineering, model training, hyperparameter tuning with Optuna, and SHAP-based explainability.
- Integrated AWS S3 for cloud artifact storage; deployed on HuggingFace Spaces with ZeroGPU for production use.

GPT-2 Language Model Trained from Scratch (124M Parameters)

PyTorch, TinyStories, Alpaca | huggingface.co/Abdourakib/tinystories-gpt2-124m

- Pre-trained a 124M parameter GPT-2 from scratch on TinyStories using PyTorch with full training pipeline including loss monitoring and checkpoint management.
- Performed three phases of Alpaca instruction fine-tuning with AdamW optimizer and cosine learning rate scheduling; published model to HuggingFace Hub.

Brain Tumor MRI Classifier - Medical Computer Vision

TensorFlow, EfficientNetB0, Grad-CAM, Streamlit | [HuggingFace Spaces](https://huggingface.co/spaces/Abdourakib/brain-tumor-mri-classifier)

- Built and deployed a CNN-based medical image classifier for brain tumor MRI scans using EfficientNetB0 with transfer learning.
- Implemented Grad-CAM visual explanations so predictions are interpretable by non-technical users; deployed as a full production application.

EDUCATION

Associate of Science, Computer Science - Community College of Philadelphia

Graduated May 2026 | Philadelphia, PA

Certifications and Training: Andrew Ng ML Specialization (Coursera) | Data Science and ML Bootcamp (Udemy, 44 hours)